

Report: 97a09efb
Generated: 2026-07-31 04:33:27Z
Run started: not recorded
Software: OriginMarker 2.2.1 (Kinetochore), Syngamy module
Sperm donor: GSM4472397_sperm_DNA_71.CEL.txt.gz (825,656 markers, SHA-256 1b56911dae7030f4)
Oocyte donor: none supplied, so a maternal origin is inferred from paternal absence rather than measured
Donor heterozygosity: 16.658% autosomal, of called markers
Samples: 1 analysed (1 androgenetic)
Assembly: GRCh37
Data handling: every file was read in the browser that produced this report. No genotype was transmitted, and none is retained.

Result summary

Sample	Origin	Absence	Ceiling	x ceiling	Zygosity	Sperm	Informative
GSM4472398_sperm_DNA_79.CEL.txt.gz	androgenetic	0.13%	1.45%	0.1x	diploid	X-bearing	608,996

Absence is the rate at which the donor's obligate allele is missing where he is homozygous. Ceiling is what this sample's own noise can manufacture: its no-call rate times its heterozygous fraction. A sample is called as lacking a paternal contribution only above 3x that ceiling, and as carrying one only at or below it; in between it is left uncalled. No confidence percentage is reported, deliberately: see Interpretation.

GSM4472398_sperm_DNA_79.CEL.txt.gz vs GSM4472397_sperm_DNA_71.CEL.txt.gz

ANDROGENETIC

Paternal genome only. No maternal complement.

absent 0.13% | ceiling 1.45% | 0.1x | 608,996 informative markers | zygosity diploid | X-bearing sperm

Evidence

Measurement	Observed	Reference	The reference is
Donor alleles absent	0.13%	1.45%	this sample's no-call rate times its heterozygous fraction. Dropout manufactures absence only by turning a heterozygous call homozygous and discarding the donor's allele, so the bound is the product and neither term alone.
Alleles the donor lacks	2.73%	8.33%	half the donor's heterozygosity. A second parent supplies an allele he lacks only where he is homozygous, at sum(pq), and his heterozygosity is sum(2pq).
Heterozygous BAF band	16.03%	8.00%	measured at 15-16% for diploid genomes and 1.3-3.4% for uniparental ones. Dropout does not mimic this: a biparental embryo at 33% dropout still reads 22.2%.

Sex chromosomes

Chr	Informative	Absent	Rate	x ceiling	Verdict	Note
chrX	18,775	388	2.07%	1.4	unclear	
chrY	781	11	1.41%	1.0	present	

Chromosomes

Chr	Informative	Absent	Rate	x ceiling	Verdict	Note
chr1	50,090	47	0.09%	0.1	present	
chr2	47,668	60	0.13%	0.1	present	
chr3	40,629	36	0.09%	0.1	present	
chr4	36,621	30	0.08%	0.1	present	
chr5	35,477	35	0.10%	0.1	present	
chr6	42,678	75	0.18%	0.1	present	
chr7	32,749	27	0.08%	0.1	present	
chr8	30,148	24	0.08%	0.1	present	
chr9	26,347	25	0.09%	0.1	present	
chrX	18,775	388	2.07%	1.4	unclear	
chrY	781	11	1.41%	1.0	present	
chr10	29,799	34	0.11%	0.1	present	
chr11	32,620	42	0.13%	0.1	present	
chr12	29,222	25	0.09%	0.1	present	
chr13	20,223	29	0.14%	0.1	present	
chr14	20,552	46	0.22%	0.2	present	
chr15	18,890	13	0.07%	0.0	present	
chr16	22,206	37	0.17%	0.1	present	
chr17	22,168	40	0.18%	0.1	present	
chr18	17,095	12	0.07%	0.0	present	
chr19	20,700	125	0.60%	0.4	present	
chr20	14,788	15	0.10%	0.1	present	
chr21	8,689	5	0.06%	0.0	present	
chr22	9,637	7	0.07%	0.1	present	

Sample quality

Markers: 825,656 read, 776,810 called (94.1%)

No-call: 5.92%
Heterozygous: 16.98% of called
chrX / autosomal het: 0.115
Sex call: male
Product: UK Biobank Axiom Array
Assembly: GRCh37 (0 placements illegal under GRCh37, 6,315 under GRCh38, 825,656 tested)
Allele coding: ok BAF corroborates 0=AA, 2=BB
Mean BAF AA/AB/BB: 0.040 / 0.507 / 0.965
SHA-256: b48d503ebc175f68d8f4d3dce66097dff58ecce8092401c0fdd31c3c4a46ac58
File size: 62,382,461 bytes

Quality gates

Gate	Value	Verdict	Basis
call rate	94.1%	usable	inside the 75-95% amplified usable band; the Axiom vendor gate of 97% would reject this sample, and that gate was established on unamplified bulk DNA
het-to-hom asymmetry valid	94.1%	usable	holds within the usable call-rate band
genome-wide LOH	17.0%	report only	no genome-wide LOH signature
heterozygosity plausible	17.0%	usable	within the plausible range for one diploid genome
numeric genotype coding	-	usable	BAF corroborates 0=AA, 2=BB
sex call	11.5%	usable	chrX/autosome heterozygosity ratio 0.115 is inside the male band. Dropout cancels in the ratio, so this survives amplified material.

Findings

Diploid but carrying 2.73% alleles the parent lacks, below the 8.33% a second parent would contribute. Consistent with a parent-only genome that is heterozygous rather than duplicated, meaning two sperm. This axis separates by about 1.6x against thirty-fold for absence, so treat a near-boundary call as provisional. Every allele traces to this parent and the genome is homozygous: a parent-only complement, duplicated. Nothing here required chrY. No chrY, yet the parent's alleles are present on chrX as well as the autosomes: an X-bearing sperm. This is the case a chrY test cannot call.

Methods

Parent of origin was determined from SNP array genotypes using OriginMarker 2.2.1 (Kinetochore). For each of 1 sample(s), the rate at which the sperm donor's obligate allele was absent was computed across autosomal markers where he is homozygous and the sample is called, and compared against the rate that sample's own genotyping noise can produce, taken as the product of its no-call rate and its heterozygous fraction. A sample was called as lacking a paternal contribution only where the observed rate exceeded that bound by 3-fold, and as carrying one only where it fell at or below the bound; rates between the two were left uncalled. Whether a second parent contributed was assessed from the rate at which the sample carries alleles the sperm donor does not possess, against half his heterozygosity, which is the contribution a second parent makes. Zygosity was taken from the fraction of B-allele frequencies falling in the heterozygous band, 0.35 to 0.65. Sperm type was inferred from the chrX rate rather than from chrY, so that an X-bearing sperm carrying a full paternal genome is distinguishable from an absent paternal contribution; neither case leaves a Y.

No oocyte donor array was supplied. A maternal origin below is therefore inferred from the absence of the paternal contribution rather than measured, and a sample belonging to neither declared parent is indistinguishable from one that is maternal in origin.

Assembly was determined from marker positions against the UCSC chromInfo and gap tables and found to be GRCh37. No liftOver was performed.

Interpretation

No confidence percentage is attached to a call, and this is deliberate. The per-sample likelihood ratio between "paternal genome present" and "absent" runs past 10^10000 on a genome-wide array, which is not a probability any reader should be handed. The honest figure is the empirical accuracy of the method, and on the pairs of known relationship it has been run against that is 9 of 9 correct, whose 95% lower bound is roughly 72%. About 300 consecutive correct calls would be needed to claim 99%. What is reported instead is the margin: the observed rate, the reference it was measured against, and the ratio between them, so a reader can see how far from the boundary the call sits.

The absence axis separates by roughly thirty-fold and carries the call. The alleles-the-donor-lacks axis separates by about 1.6-fold and only distinguishes a biparental genome from a paternal-only one that is heterozygous; treat a near-boundary call on that axis as provisional. A sample scoring in the uncalled band is not a weak positive, it is an absence of evidence: an array from the second parent measures dropout directly and settles it.

Constants and where they come from

Constant	Value	Provenance
Mendelian inconsistency floor	0.0063	Kothiyal 2019, 0.63% of variant sites across 1,314 nuclear families. A lower bound on the spurious-violation rate, never the value itself.
Paternal-absence noise bound	no-call rate x heterozygous fraction	Derived per sample, not fitted. Bounds every related pair measured, within 2x: 0.13% predicted against 0.05% observed, 0.18% against 0.16%, 10.4% against 9.69%.
Absence calling margin	3x	How far above the noise bound an absence must sit before it is called.
Residual absence floor	0.005	Absence on clean data from genotyping error alone, measured at 0.03% and 0.05%. Added to the bound so a near-perfect array still has a non-zero reference.
Second-parent signal	half the donor's heterozygosity	Derived. A second parent supplies an allele the donor lacks only where he is homozygous, at sum(pq); his heterozygosity is sum(2pq) over the same markers.
Diploid heterozygous BAF band	0.08 of markers in 0.35-0.65	Measured at 15-16% for diploid genomes and 1.3-3.4% for uniparental ones. Dropout does not mimic this: a biparental embryo at 33% dropout still reads 22.2%.
Per-chromosome minimum	200 informative markers	Below this a chromosome is not reported: the rate is too noisy to place against the ceiling.
Assembly tables	UCSC chromInfo and gap, hg19 and hg38	Freely redistributable. No liftOver chain file is used or shipped: those carry a non-commercial field-of-use restriction Apache 2.0 cannot sublicense.

Files read

File	Role	Bytes	Markers	SHA-256
GSM4472397_sperm_DNA_71.CEL.txt.gz	donor	63,674,062	825,656	1b56911dae7030f4ecec2ade9c7b1bacee23def81d130e6b3d0ac9edb388169
GSM4472398_sperm_DNA_79.CEL.txt.gz	sample	62,382,461	825,656	b48d503ebc175f68d8f4d3dce66097dff58ecce8092401c0fdd31c3c4a46ac58

The hash is of the file as read, so a reviewer can confirm the input without being sent it.

Citation

OriginMarker 2.2.1 (Kinetochore). Syngamy: parent of origin from SNP array genotypes. Report 97a09efb, generated 2026-07-31 04:33:27Z. Kothiyal P et al. Mendelian inconsistent signatures from 1,314 ancestrally diverse family trios. J Comput Biol 2019;26(5):405-419. doi:10.1089/cmb.2018.0253. Cited for the inconsistency floor above.